ASSIGNMENT

Introduction to Asymmetric Cryptography

Subject: Cryptography and Network Security

Q1. Explain the principles of asymmetric cryptography, public/private keys, confidentiality, authentication and non-repudiation.

Asymmetric (public-key) cryptography uses a mathematically related pair of keys: a public key that can be shared and a private key that must be kept secret. For confidentiality, Alice encrypts a message using Bob's public key, and only Bob's private key can decrypt it. For authentication and digital signatures, Alice signs data using her private key and others verify the signature using Alice's public key. A valid digital signature can support integrity, authentication and non-repudiation, provided the private key is securely controlled and Alice's public key is reliably linked to her identity.

Diagram:
Alice: Plaintext → Encrypt with Bob's public key → Ciphertext → Network → Decrypt with Bob's private key → Bob: Plaintext
For a signature: Alice signs with Alice's private key → Signature; Bob verifies with Alice's public key.

Q2. Differentiate between symmetric and asymmetric cryptography.

Symmetric cryptography uses one shared secret key for encryption and decryption. Asymmetric cryptography uses a public/private key pair. Symmetric algorithms are usually faster and suitable for bulk data, while asymmetric algorithms are slower and commonly used for key establishment, authentication and digital signatures.

Q3. Describe trapdoor one-way functions, Diffie-Hellman, digital signatures and PKI with certificates.

A trapdoor one-way function is easy to compute in one direction but difficult to reverse unless special secret information (the trapdoor) is known. RSA is based on a related mathematical idea: operations are easy with the relevant key, while recovering the private information from the public modulus is believed to be difficult for appropriately chosen parameters.

Diffie-Hellman key exchange allows two parties to derive a shared secret over a public channel. By itself, unauthenticated Diffie-Hellman does not prove who the other party is and can be vulnerable to a man-in-the-middle attack; authentication is needed.

Digital signatures are generated with a private signing key and checked with a public verification key. They help verify the signer and detect message changes, and can support non-repudiation when identity and key control are trustworthy.

Public-key infrastructure (PKI) manages public-key certificates and trust relationships. A certificate authority (CA) signs a digital certificate that binds a public key to an identity. A relying party checks the certificate chain, validity period, intended use and revocation status as applicable.

Q4. Trace the historical development of public-key cryptography.

Public-key cryptography addressed the key-distribution problem: in a symmetric system, communicating parties need to share a secret key securely before protected communication. In 1976, Whitfield Diffie and Martin Hellman published a landmark paper describing public-key ideas and a method for establishing a shared secret over an insecure channel. In 1977, Ronald Rivest, Adi Shamir and Leonard Adleman introduced RSA, which supports encryption schemes and digital signatures when used with appropriate padding and protocols.

Earlier related work had been conducted secretly at the UK Government Communications Headquarters (GCHQ). James Ellis developed the concept of non-secret encryption, Clifford Cocks devised an RSA-like system, and Malcolm Williamson developed a key-exchange method. Their work was classified and became publicly known later. ElGamal cryptography was introduced in the 1980s and is based on the discrete logarithm problem. Elliptic Curve Cryptography (ECC) uses elliptic-curve mathematics to provide comparable security with much smaller keys than traditional RSA keys.

Q5. Discuss advantages and disadvantages of asymmetric cryptography and explain hybrid systems such as TLS.

Advantages: public keys can be distributed openly; users do not need a separate shared secret with every other user; digital signatures support authentication and integrity; and public-key methods help establish session keys across untrusted networks.

Disadvantages: public-key operations are slower than symmetric encryption; certificates and private keys need careful management; implementations can be vulnerable to side-channel attacks; and public keys must be authenticated to prevent impersonation.

Real systems therefore use hybrid cryptography. For example, TLS authenticates the server using a certificate and uses a key-establishment mechanism to derive shared session keys. The actual application data is then protected with fast symmetric authenticated encryption, such as AES-GCM or ChaCha20-Poly1305. This combines practical key establishment and authentication with efficient data protection.

Q6. Describe RSA key generation, encryption and decryption with formulas and flowchart.

RSA key generation:
1. Choose two large distinct primes p and q.
2. Compute n = p × q. The modulus n is part of both the public and private key.
3. Compute $\phi(n) = (p - 1)(q - 1)$.
4. Choose a public exponent e such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$; a common choice is 65537.
5. Compute the private exponent d as the modular inverse of e: $d \times e \equiv 1 \pmod{\phi(n)}$.
6. Public key: (n, e). Private key: d together with the required private parameters.

Encryption: represent the properly encoded message as an integer m with $0 \leq m < n$. Compute $c = m^e \bmod n$.
Decryption: compute $m = c^d \bmod n$, then decode the recovered message.

Flowchart:
Choose p, q → Compute n and $\phi(n)$ → Choose e → Compute d → Publish (n,e), protect d → Encode message and compute $c = m^e \bmod n$ → Send c → Recover $m = c^d \bmod n$ → Decode message.

Decryption works because $ed \equiv 1 \pmod{\phi(n)}$; under RSA's mathematical conditions, modular exponentiation with d reverses exponentiation with e. In practice, secure RSA encryption uses a standardized encoding/padding scheme such as OAEP; raw textbook RSA must not be used.

Q7. Explain RSA key size, factoring, recommended sizes, ECC comparison and quantum threats.

RSA key size is the bit length of the modulus n. RSA security relies largely on the difficulty of factoring n into its prime factors p and q; if an attacker factors n, they can derive the private key. A 1024-bit RSA key is considered too weak for new security deployments. 2048-bit RSA is a common minimum for many current applications, while 3072-bit RSA offers a higher security margin and is often compared with roughly 256-bit ECC. 4096-bit RSA provides a larger margin but requires more computation. Approximate comparisons vary by standard and use case; ECC can achieve similar classical security with much shorter keys.

A sufficiently capable fault-tolerant quantum computer running Shor's algorithm could factor RSA moduli efficiently and break RSA. Large-scale quantum computers of that capability do not currently exist, but long-lived systems should plan migration to standardized post-quantum cryptography.

Q8. Discuss RSA attacks, padding, signatures and good deployment practices.

Factoring attacks: factoring the public modulus can reveal the private key, so use adequately sized keys generated with secure randomness.

Small public exponent and small message attacks: using a small exponent without secure randomized padding, especially with poorly structured or repeated messages, can expose plaintext. Use standardized RSA-OAEP for encryption and avoid raw RSA.

Common modulus attack: reusing the same modulus with different public exponents can create vulnerabilities in certain conditions. Each key pair should be generated and managed appropriately rather than sharing a modulus across users.

Timing and side-channel attacks: execution time, power use, cache behavior or other physical signals may leak private-key information. Use vetted cryptographic libraries with constant-time implementations and appropriate protections.

Chosen-ciphertext attacks: an attacker may submit selected ciphertexts and use decryption behavior or error differences to learn information. OAEP and carefully designed protocols help reduce these risks; error handling should not reveal sensitive details.

Digital signatures: a signer uses the private key to create a signature over a message digest with a secure signature scheme such as RSA-PSS. The recipient verifies it using the public key. Signatures support integrity and authentication and may support non-repudiation when identity, key custody and procedures are reliable.

Good practices: use at least 2048-bit RSA for applicable current uses; prefer 3072-bit keys when a longer security margin is needed; use OAEP for encryption and PSS for signatures; generate keys with a cryptographically secure random generator; protect private keys with access controls or hardware security modules; validate certificates and key purposes; use maintained libraries; avoid custom cryptography and textbook RSA; rotate or revoke compromised keys; and plan for post-quantum migration.

References

1. William Stallings, Cryptography and Network Security: Principles and Practice.

2. NIST publications on RSA, digital signatures and public-key cryptography.

3. Whitfield Diffie and Martin E. Hellman, New Directions in Cryptography, 1976.